

Letter from the Editor-in-Chief

Welcome to the Summer 2026 issue of the International Journal of Secondary Computing and Applications Research (IJSCAR). This second issue of Volume 3 brings together five original research papers representing a wide spectrum of computing subfields — from reinforcement learning and robotics to thermodynamics-inspired generative models, cybersecurity, and AI policy analysis. Each paper reflects the intellectual ambition and research rigor we have come to expect from IJSCAR authors.

This issue opens with a study of hyperparameter sensitivity in two-dimensional drone navigation, in which the author systematically investigates how reward-shaping strategies and discount factors affect agent performance in a stochastic environment — a small-scale investigation with meaningful implications for real-world UAV systems. The second paper applies non-equilibrium thermodynamics to the design of noise schedules for denoising diffusion probabilistic models, deriving an entropy-minimal schedule from first principles and validating it against standard baselines. The third contribution presents Plexibot, a homogeneous modular robot framework capable of both aerial and terrestrial locomotion, with a novel docking mechanism that enables decentralized, consensus-based reconfiguration across unstructured environments.

The fourth paper turns to human-computer interaction and governance, offering a systematic heuristic evaluation of twenty U.S. municipal AI policy documents alongside an NLP-based readability analysis. The findings reveal a striking infrastructure-interface gap: cities invest in governance structure but consistently underperform on plain-language accessibility and actionability — a consequential finding as AI governance scales. The issue closes with an investigation into backdoor attacks on reinforcement learning agents controlling electric vehicle charging systems, demonstrating that trajectory-level behavioral monitoring can detect compromised agents while also surfacing the limits of current detection methods under adversarially adaptive conditions.

Taken together, the papers in this issue reflect the genuine diversity of secondary computing research: theoretical, applied, experimental, and evaluative work appearing side by side. As Editor-in-Chief, I remain grateful to our authors for their commitment to the research process, to our reviewers for their careful and constructive feedback, and to the mentors and educators who make this work possible.

Sincerely,

Maria Hwang

Editor-in-Chief

